LAB REPORT

11th March 2025

Data Recovery Practical

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Session : 11th February – 22nd April 2025

For the class of Computer Forensic, taught by

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Contents

1 Introduction

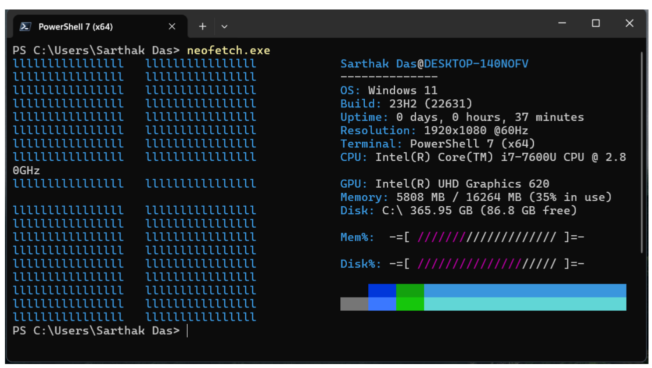
1.1 Objectives

This report is a documentation of my lab work as assigned on 4th March, 2025. The primary objectives of this lab are three-fold:

- Recovering Data from a Specific Partition
- Recovering Data from a Lost Partition
- Recovering Data from a Merged Partition

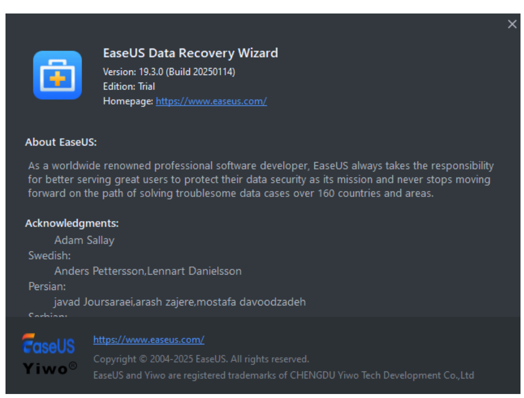
1.2 System Information

For the purposes of this lab, a Lenovo ThinkPad X270 was used, having the following system specifications: Intel Core i7-7600U CPU @ 2.80GHz with Intel UHD Graphics 620, 16GB RAM, 1TB SSD running operating system Windows 11 23H2 (Build 22631).



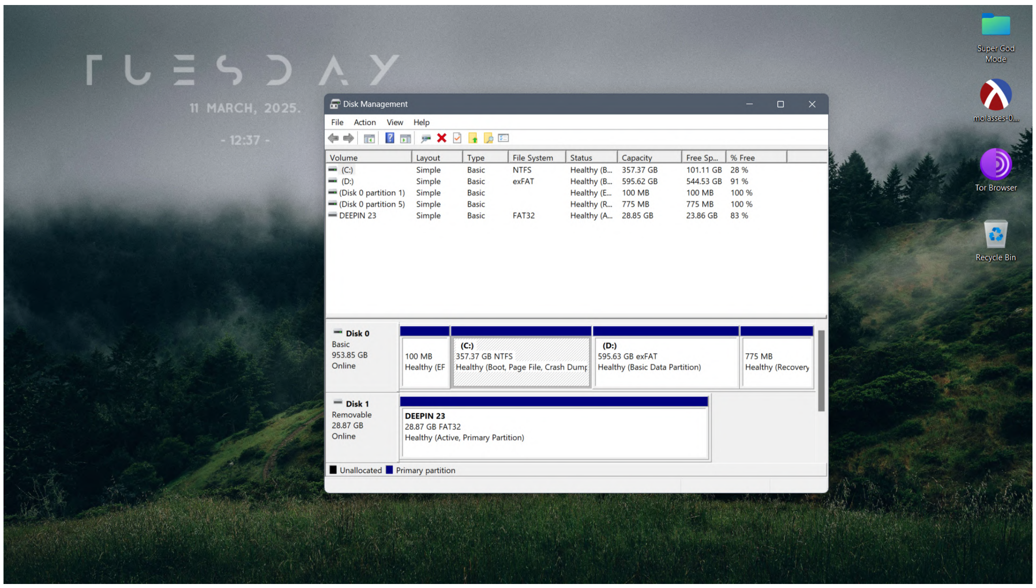
1.3 Software Used

We use EaseUS Data Recovery Wizard (Trial Edition) version 19.3.0 build 20250114.

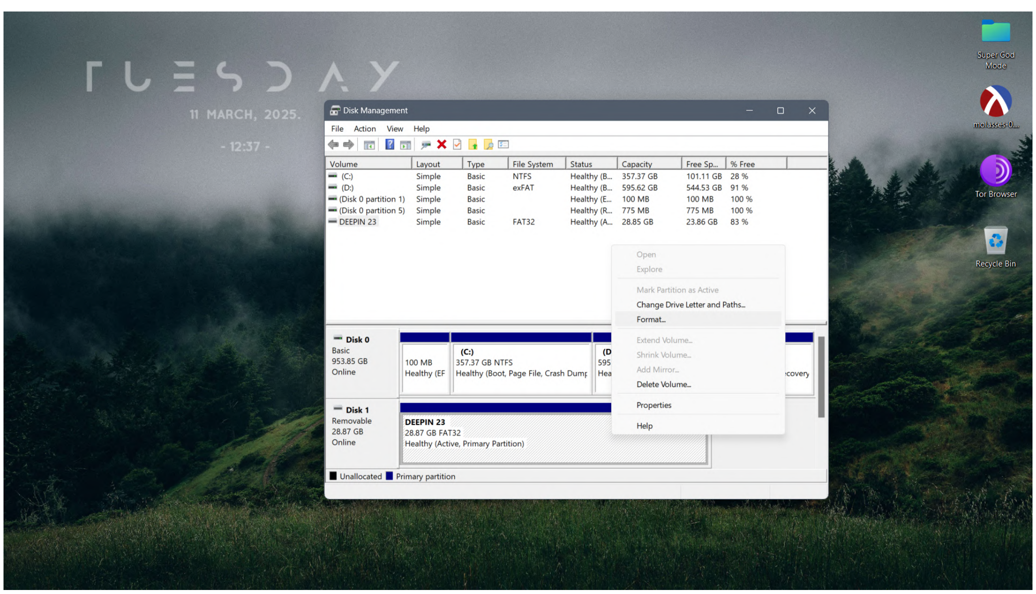


2 Objective 1: Data Recovery from Specific Partition

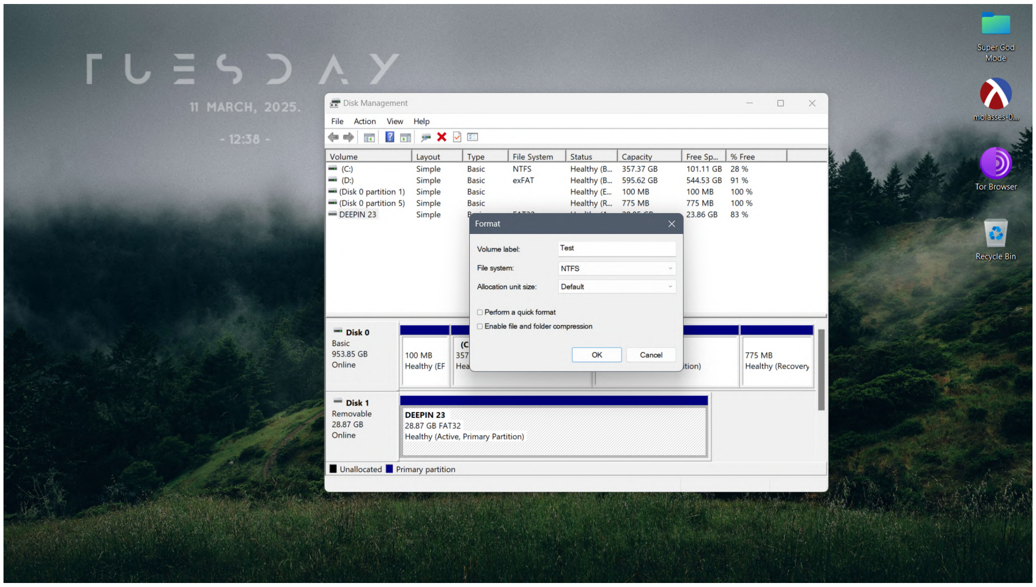
We begin by going to the Start Menu, typing "Disk Management" and selecting the *Create and format hard disk partitions* option. This opens up the **Disk Management Console**.



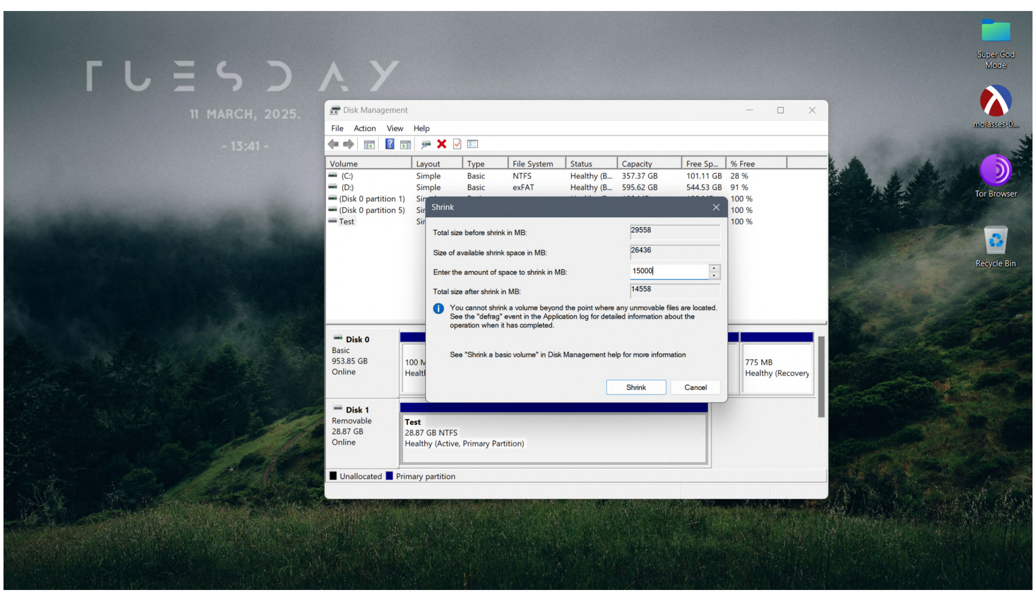
For the purposes of this lab, we have selected a 32GB Kingston DataTraveler 3.0 USB stick as our storage drive. We locate the drive on the console, right click and select *Format*. We confirm our choice by clicking "Yes".



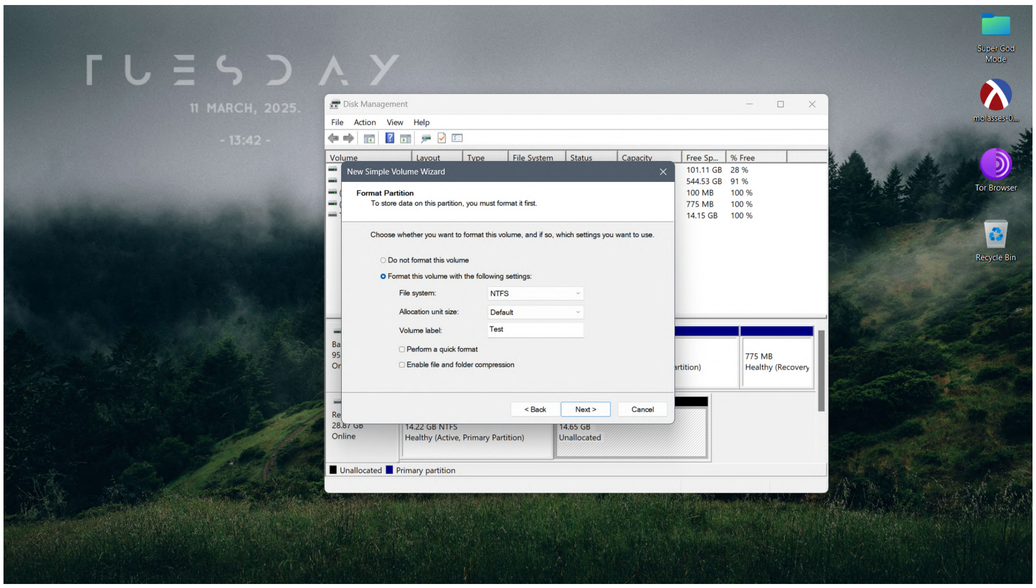
We proceed to format the drive into NTFS format, and make sure that the quick format option is de-selected. This ensures that a full format occurs and the drive is wiped sector by sector, allowing us to start afresh.



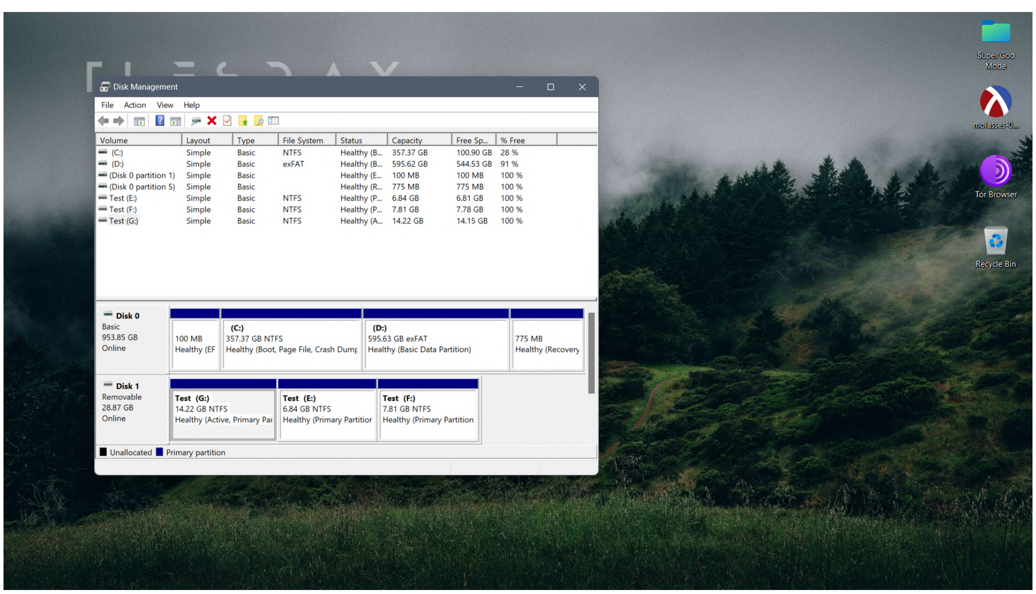
Once formatted, we select the drive, right click and select *Shrink Volume*, and shrink the volume size to 14.22GB.



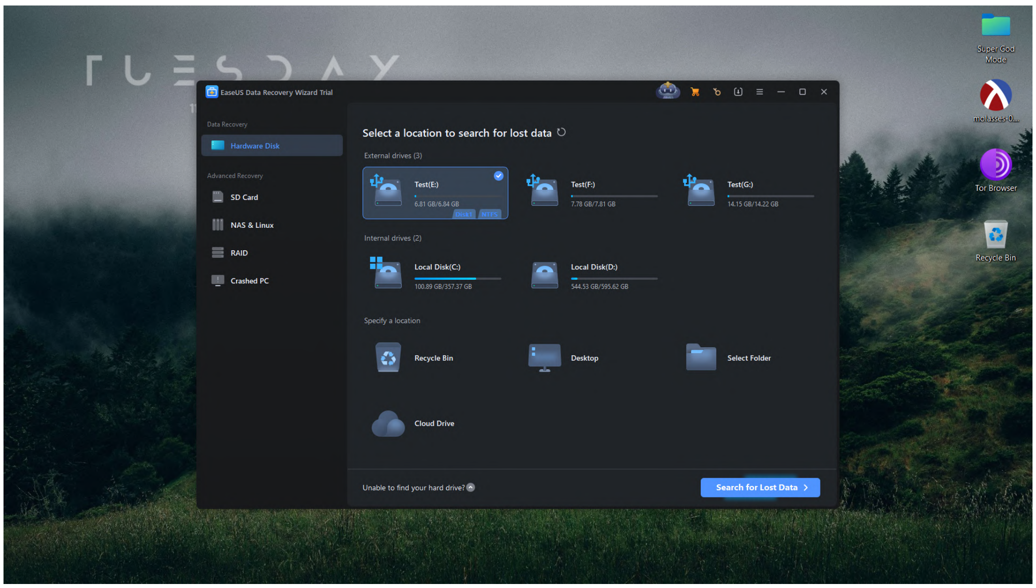
For the unallocated space, we right click and select *New Simple Volume*. This opens up the **Simple Volume Wizard**. We set the volume size to 6.84GB, the drive format to NTFS, and perform a full format.



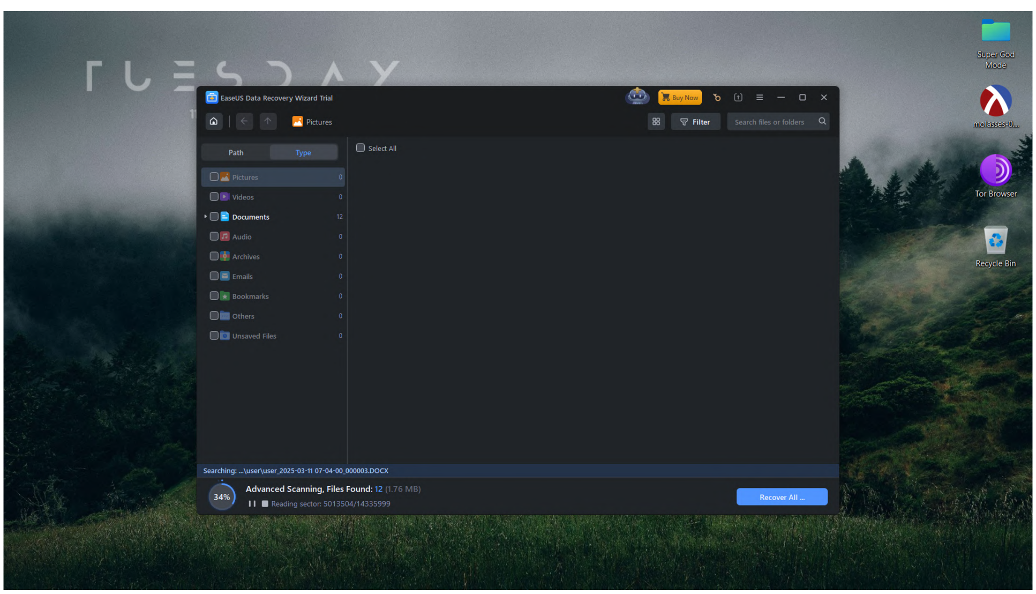
Repeating the process for the remaining 7.81GB unallocated space, we now have 3 fresh, fully wiped NTFS partitions in our test drive. Note that the first partition cannot be accessed yet as it was created via a full format of an existing physical drive. In order to mount it, we select the partition, right click and select *Change Drive Letter and Paths* and add a drive letter.



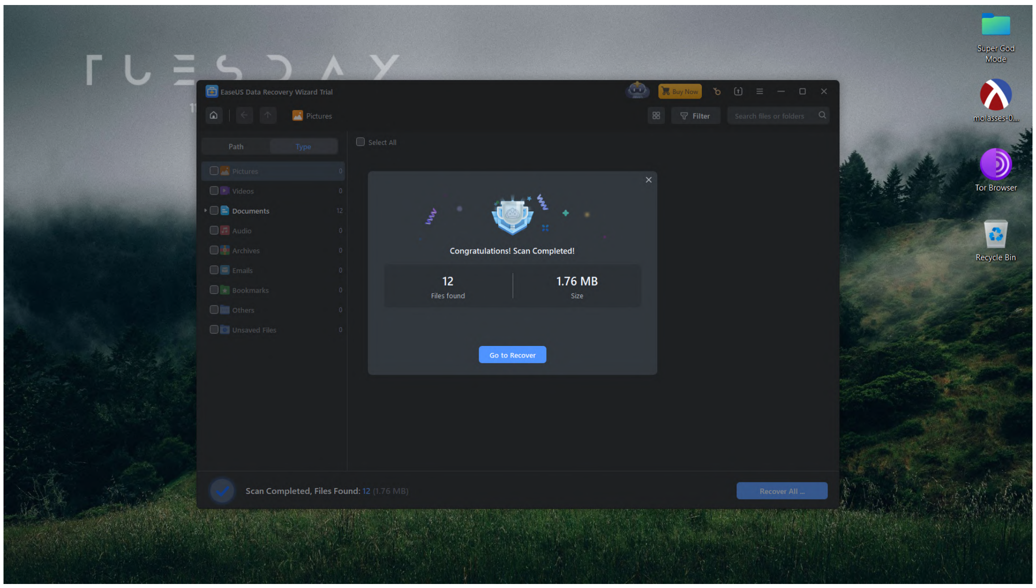
We copy some test data into the any of the three partitions, delete that data, and empty the Recycle Bin. We can now begin the data recovery. We begin by opening the EaseUS Data Recovery Wizard. We select the partition where we had placed our test data, and click *Search for Lost Data*.



EaseUS Data Recovery Wizard scans for any recoverable data.

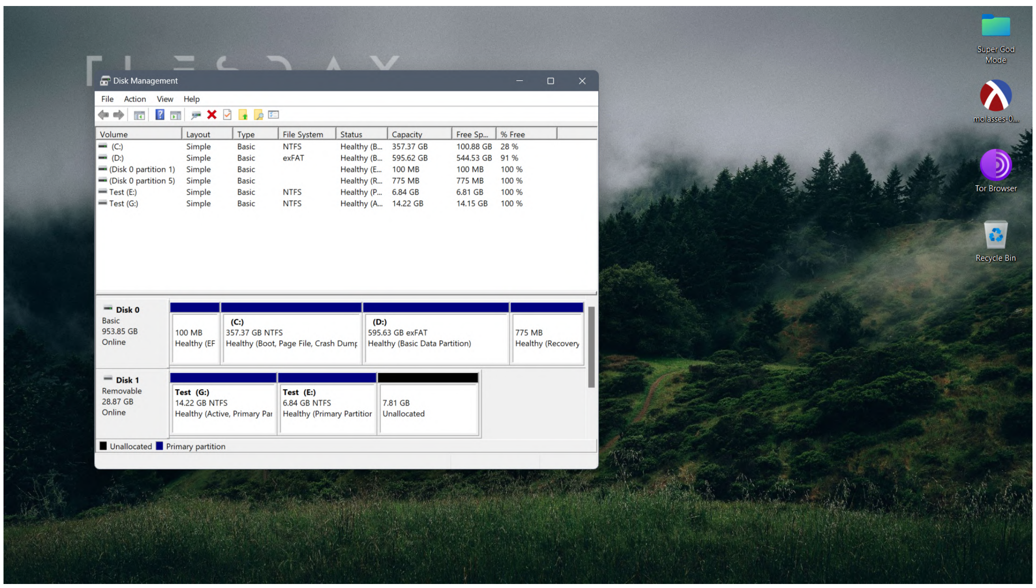


As we can see, EaseUS Data Recovery Wizard has found 12 files of data worth 1.76MB, which is our deleted data. We can click *Recover All* to begin the recovery process. Please note that we are using the Trial Edition, hence, data recovery is not supported.

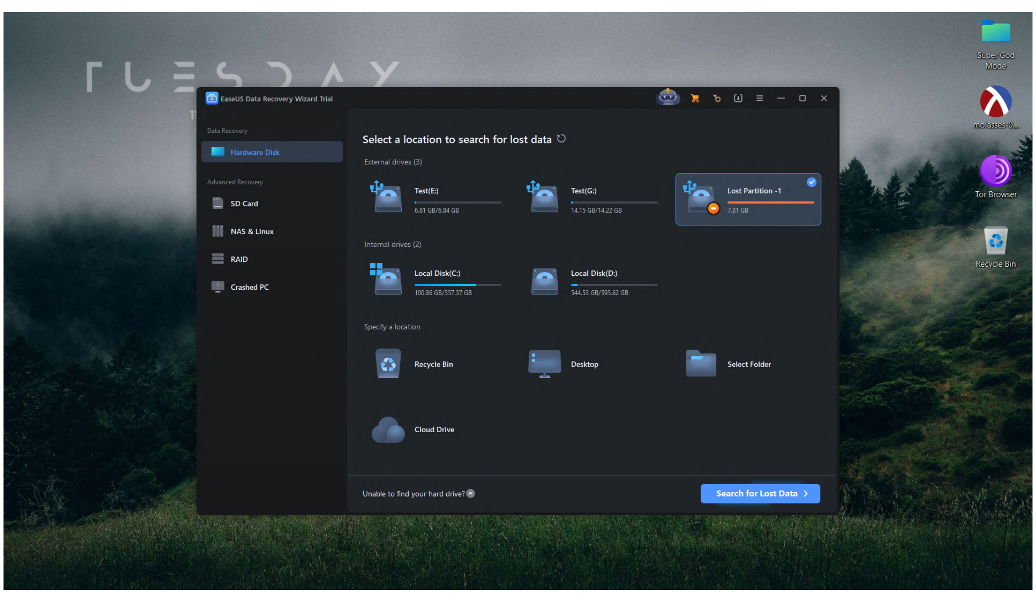


3 Objective 2: Data Recovery from Lost Partition

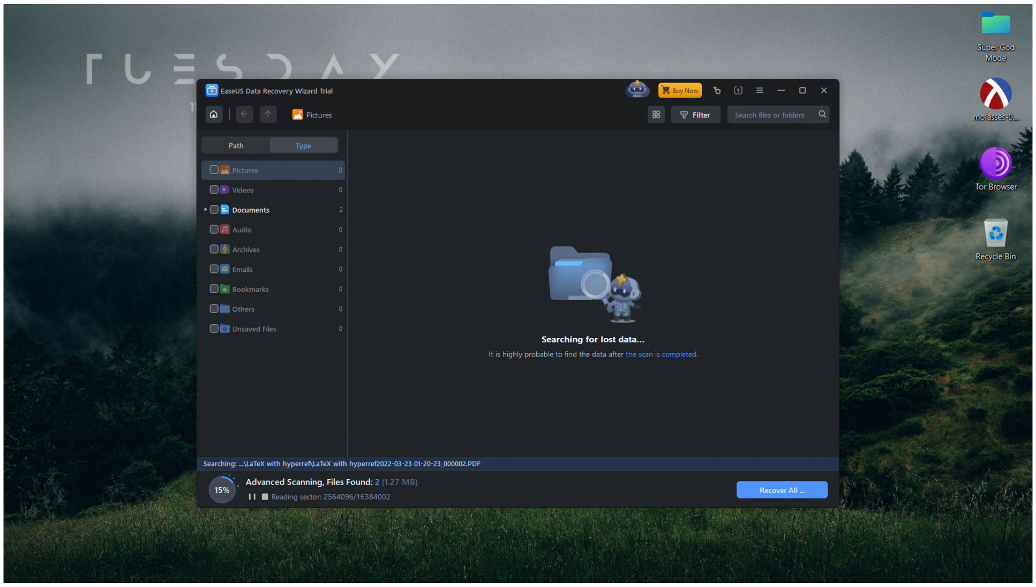
For the next objective, we first copy a fresh test data into the any of the remaining two partitions. Then, we open the **Disk Management Console** once again and select that partition. We then right click and select *Delete Volume*. This unallocates the space and our partition is lost, virtually deleting our data.



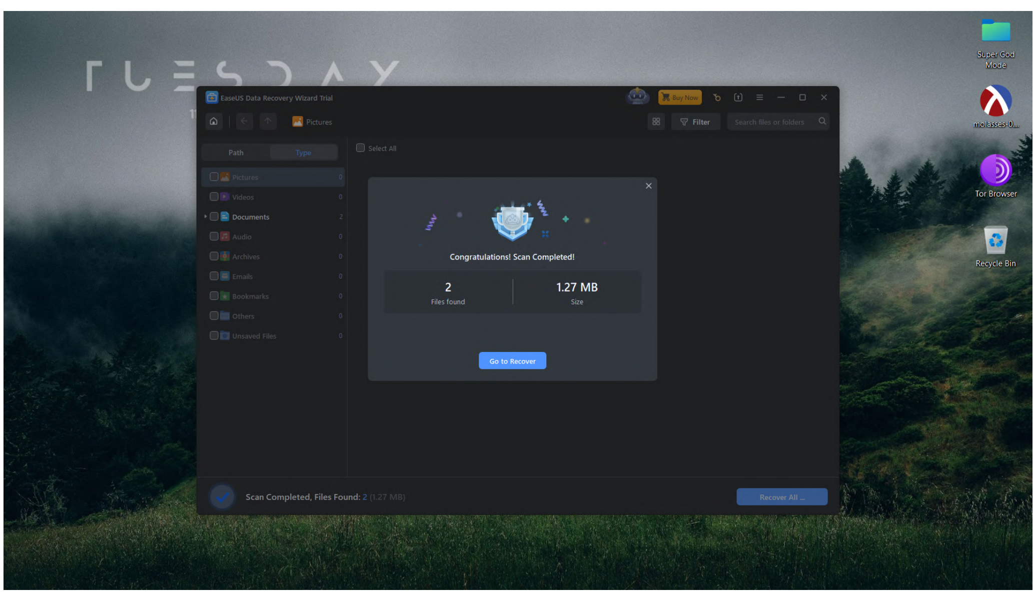
We open the EaseUS Data Recovery Wizard. The wizard detects our lost partition. We select the lost partition and click *Search for Lost Data*.



EaseUS Data Recovery Wizard scans for any recoverable data.

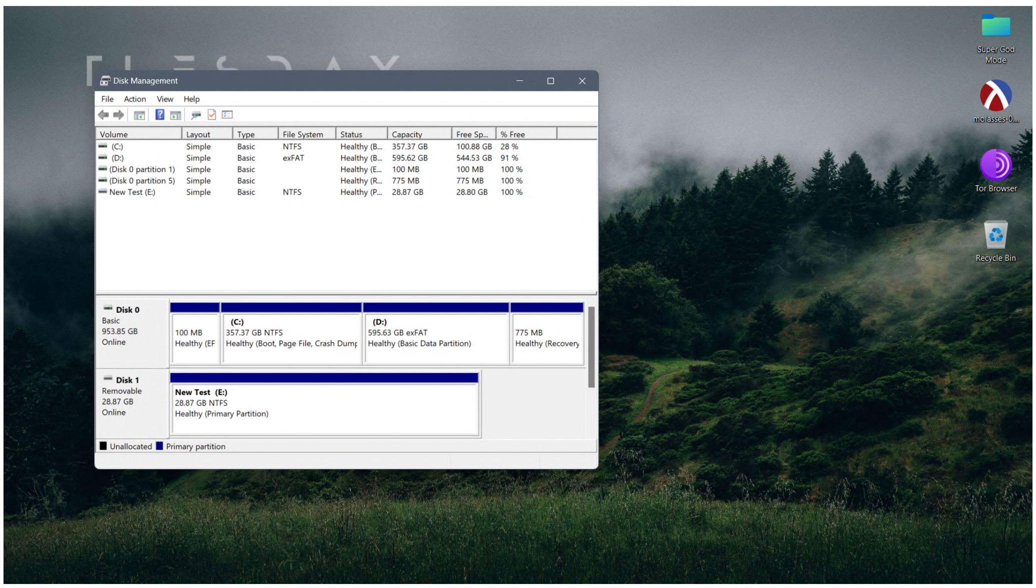


As we can see, EaseUS Data Recovery Wizard has found 2 files of data worth 1.27MB, which is the data we had copied to the lost partition. We can click *Recover All* to begin the recovery process. Please note that we are using the Trial Edition, hence, data recovery is not supported.

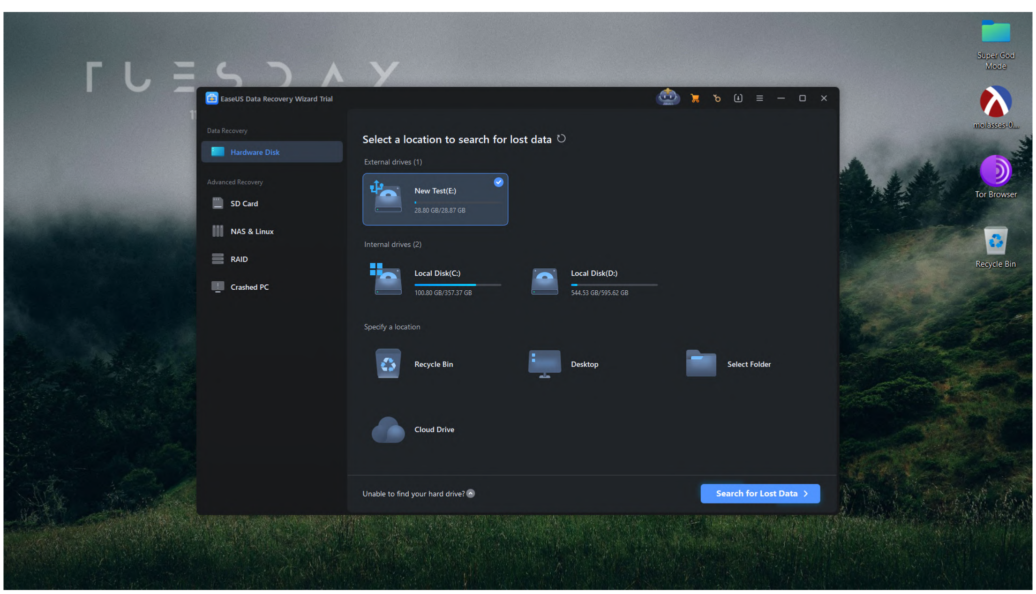


4 Objective 3: Data Recovery from Merged Partition

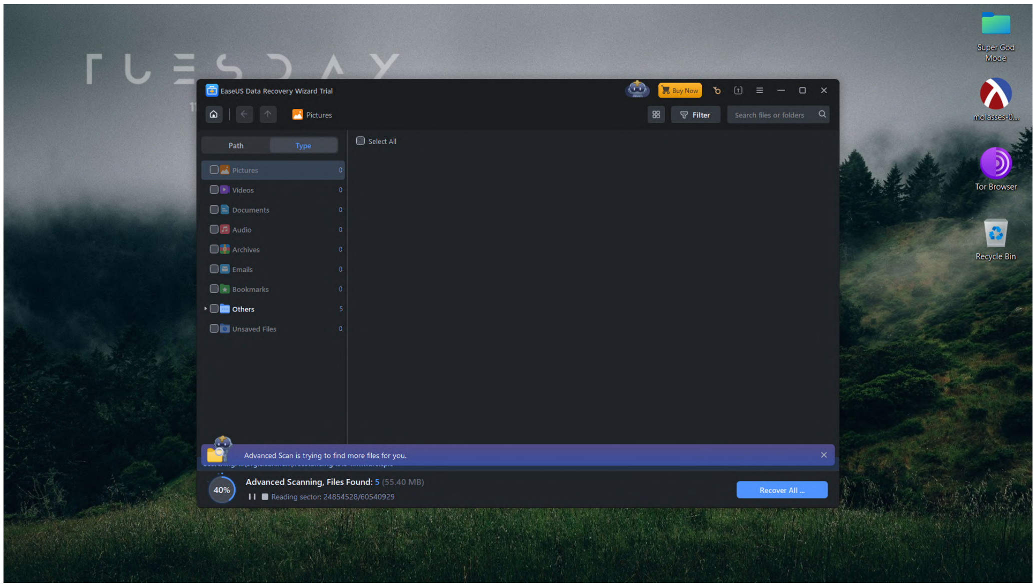
For the final objective, we first copy a fresh test data into the any of the remaining allocated partitions. Then, we open the **Disk Management Console** once again and delete all volumes. We select the unallocated space, right click and select *New Simple Volume*. This opens up the **Simple Volume Wizard**. We set the volume size to its full capacity, the drive format to NTFS, and perform a quick format. This merges our partitions into a single one.



We open the EaseUS Data Recovery Wizard, select the merged partition, and click *Search for Lost Data.*



EaseUS Data Recovery Wizard scans for any recoverable data. The wizard defaults to advanced scan for the merged partition as indicated on the screen.



As we can see, EaseUS Data Recovery Wizard has found 14 files of data worth 82.29MB, and reconstructed all the lost partitions. We can click *Recover All* to begin the recovery process. Please note that we are using the Trial Edition, hence, data recovery is not supported.